

Syed Mustafa Mohiuddin

Mechatronics Engineer — Automation, Control Systems & Electrical Design

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[Portfolio](#)



PROFESSIONAL SUMMARY

Mechatronics engineer with direct experience developing and commissioning automated X-ray imaging systems, designing motion control and electrical systems, and programming hardware-software interfaces for high-precision industrial machines. Proven ability to design circuit diagrams and wiring harnesses, perform risk and safety analysis against IEC/ISO standards, and create technical concepts and specifications for complex electromechanical systems. Experienced collaborating across interdisciplinary, international engineering teams, interpreting customer and production requirements, and proactively resolving technical issues before escalation.

ENGINEERING EXPERIENCE

Siemens Healthineers AG, Kemnath

Working Student – Automation & Control Systems Engineering

March 2025 – Present

- Optimised hardware control algorithms for automated X-ray CT imaging systems, tuning motor drive parameters across horizontal and vertical motion axes to bring system response within defined machine tolerances.
- Developed and implemented automated test programmes using Drive Manager and TCB software, replacing manual verification routines with scripted, repeatable measurement sequences for multi-series machine qualification.
- Participated in risk and safety analysis on automated imaging systems, verifying electromechanical safety parameters under full operating load and documenting compliance evidence against machine safety standards.
- Collaborated across mechanical, electronics, and software sub-teams in an international engineering environment, interpreting technical requirements and proactively resolving design-to-build integration issues before escalation.

Siemens Healthineers AG, Kemnath

Bachelor Thesis – Automated Motion Control System

October 2024 – February 2025

- Designed and implemented a dynamic control system for an automated X-ray patient positioning platform, programming the hardware-software interface to regulate motor torque based on real-time load feedback.
- Developed control logic in MATLAB/Simulink and validated via Hardware-in-the-Loop (HIL) testing, identifying and resolving edge cases before physical system commissioning.
- Authored full technical documentation: system concept, control architecture, test specifications, and risk analysis to IEC 60601-2-44 and ISO 15066 standards – from concept through series production handover.

CORE COMPETENCIES

Automation & Control Systems

Hardware-software interface development; motion system design and commissioning; PLC/embedded control logic; machine control optimisation for automated inspection systems.

Electrical Design

Circuit diagram creation; industrial wiring harness and control panel design; power distribution systems; EMC and insulation compliance.

X-ray & Imaging Systems

Direct experience on automated X-ray CT imaging platforms: motion control, sensor integration, safety interlock systems, and machine commissioning.

Risk & Safety Analysis

Machine safety analysis; IEC 60601-2-44, ISO 15066 compliance; regulatory documentation and safety evidence packages.

Technical Documentation

System concepts, specifications, circuit diagrams, test protocols, risk analyses, and handover documentation to production-ready standards.

TECHNICAL SKILLS

Automation & Control

PLC programming (CoDeSys-compatible embedded controllers), motion control, drive systems, hardware-software interfacing, MATLAB/Simulink, HIL testing

Siemens Healthineers AG, Kemnath

Mechatronics Intern – X-ray Imaging Systems & Electrical Design

April 2024 – September 2024

- Engineered a safety interlock and collision avoidance system for a motorised X-ray CT gantry, integrating SICK LiDAR sensors and programming the hardware-software interface for automated obstacle detection.
- Designed and created circuit diagrams, wiring harnesses, and power distribution schematics for multiple X-ray imaging system prototypes, ensuring EMC and insulation compliance throughout.
- Commissioned motion systems for robotic imaging platforms from circuit diagram to full operational validation; installed sensors, actuators, and drive motors to specification and conducted structured commissioning tests.
- Interpreted customer and production requirements working closely with product management teams; created technical concepts and specifications, and supported service and applications teams in resolving hardware integration issues.

Kraken Robotics, Rostock

Electrical Engineering Intern – Autonomous Systems

October 2023 – March 2024

- Maintained and diagnosed automated underwater robot platforms, applying structured fault-isolation methodology to electrical and sensor hardware to ensure machine reliability between operational deployments.
- Supported field service and applications efforts on autonomous systems, diagnosing and resolving hardware faults under operational conditions and communicating technical solutions to project teams.

EDUCATION

Rhine-Waal University, Kleve — *M.Sc. Engineering in Bionics*

Expected March 2027

Adaptive systems, control theory, sensors, sustainable engineering

Rhine-Waal University, Kleve — *B.Sc. Mechatronics Systems Engineering*

February 2026

Control Systems, Automation, Sensors & Actuators, Microcontrollers, Robotics

Cambridge Int. Examinations, Karachi — *A-Levels — Abitur Grade 1.8*

2020

O-Levels: 1A*, 7A

KEY ENGINEERING PROJECTS

X-ray Gantry Collision Avoidance System — *Siemens Healthineers*

Designed and commissioned a safety interlock automation system for a motorised X-ray CT imaging gantry: programmed hardware-software interface logic, created all circuit diagrams, mechanical mounting designs, and cable routing from concept to validated installation. Directly comparable to AXI machine control development.

Automated Motion Control – Electrical Design & Commissioning — *Technical Lead*

Designed a full electromechanical control system for automated X-ray patient positioning, including control logic development, circuit diagram creation, wiring harness design, HiL testing, and risk analysis against IEC/ISO machine safety standards. Delivered complete technical concept and specification documentation.

Electrical & CAD Design

Circuit diagram design, SolidWorks, Siemens NX, AutoCAD, Fusion 360, wiring harness design, PCB layout (EAGLE, KiCad, Altium)

Standards & Compliance

IEC 60601-2-44, ISO 15066, ISO machinery safety standards, risk and safety analysis, regulatory compliance documentation

Programming

C/C++, Python, MATLAB/Simulink, STM32, Arduino, Raspberry Pi, MS Office

Hardware

Sensor integration (LiDAR, IMU, cameras), motor drive systems, PCB handling, soldering, cable harnessing, circuit troubleshooting

CERTIFICATIONS & AWARDS

Winner – Innovation Think Tank (ITT)

Siemens Healthineers, Bengaluru – international engineering jury.

Siemens NX – Professional Certification

SolidWorks – Mechanical Design Associate

MATLAB – Simulink Onramp

AutoCAD • Altium • Additive Manufacturing

LANGUAGES

English C1 – Fluent (written & spoken)

German B1 – Conversational & technical

Urdu / Hindi C2 – Native

OTHER

Head Delegate – HNMUN 2023

Harvard World MUN – led multinational team; international communication and negotiation.

Travel: Open to 15%+ travel

Driver's Licence: In progress (May 2026)

REFERENCES

Ebersberger, Lucas

Head of Systems Engineering

Siemens Healthineers AG

Contact details available upon request.